

Homework 4

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GitHub Repository: <https://github.com/Gilbertofeliu/ECGR-5106-HW4>

All results are generated from executed notebooks and the reproducible PyTorch script in `src/hw4_transformers.py`.

Problem 1 Analysis

The best paragraph transformer accuracy was 0.3766 at sequence length 20. The best Homework 2 RNN-family model was GRU at sequence length 10 with validation accuracy 0.5348. The transformer used more parameters and showed the expected increase in approximate attention cost as sequence length increased.

Problem 2 Analysis

The lowest Tiny Shakespeare validation loss was 2.1666 for L=4, H=4, seq=30. Deeper models improved loss but increased parameter count and training time. More heads were not automatically better because the embedding dimension was fixed.

Problem 3 Analysis

The best English-to-French transformer BLEU-4 was 0.0067. The Homework 3 attention GRU remained stronger, which is expected because the transformer run was deliberately compact and short on CPU. The samples and loss curves show partial vocabulary learning rather than full convergence.

Problem 4 Analysis

The best French-to-English transformer BLEU-4 was 0.0118. This direction had lower validation loss than English-to-French, most likely because the English target vocabulary is smaller and easier to predict in this dataset.

Problem 1 Transformer Results

sequence length	train loss	valid loss	valid accuracy	valid top3 accuracy	valid perplexity	parameter count	approx attention ops	train seconds
10	1.9402	2.1956	0.3752	0.6173	8.9855	61485	562560	1.2830
20	1.9104	2.2132	0.3766	0.6069	9.1451	61965	1144320	2.1823
30	1.9969	2.2845	0.3473	0.5836	9.8211	62445	1745280	3.2190

Problem 1 Homework 2 RNN Comparison

cell type	sequence length	valid loss	valid accuracy	valid top3 accuracy	valid perplexity	parameter count	train seconds
RNN	10	3.3245	0.4875	0.6685	27.7859	18285	3.8684
LSTM	10	3.3237	0.4262	0.6490	27.7635	55725	3.6787
GRU	10	2.8331	0.5348	0.6741	16.9983	43245	3.0301
RNN	20	3.3159	0.5014	0.6462	27.5462	18285	3.5116
LSTM	20	2.8391	0.4930	0.6518	17.1002	55725	3.1403
GRU	20	3.3166	0.5153	0.6685	27.5666	43245	4.0103
RNN	30	3.2631	0.4735	0.6518	26.1316	18285	3.1990
LSTM	30	2.9838	0.4986	0.6630	19.7635	55725	3.4524
GRU	30	3.1755	0.4986	0.6574	23.9399	43245	3.6340

Problem 2 Transformer Results

experiment	sequence length	num layers	num heads	valid loss	valid accuracy	valid perplexity	parameter count	train seconds
problem2_baseline	20	2	2	2.1842	0.3748	8.8837	63032	2.9706
problem2_baseline	30	2	2	2.1894	0.3739	8.9302	63512	4.3656
problem2_hyper	30	1	2	2.2501	0.3607	9.4890	35240	2.3516
problem2_hyper	30	1	4	2.2880	0.3406	9.8555	35240	2.6804
problem2_hyper	30	2	2	2.1894	0.3739	8.9302	63512	4.3874
problem2_hyper	30	2	4	2.2272	0.3655	9.2742	63512	4.9727
problem2_hyper	30	4	2	2.1705	0.3830	8.7623	120056	8.2298
problem2_hyper	30	4	4	2.1666	0.3732	8.7290	120056	9.4246
problem2_seq50	50	2	2	2.2442	0.3584	9.4330	64472	7.6010

Problem 2 Homework 2 RNN Comparison

experiment	cell type	sequence length	valid loss	valid accuracy	valid perplexity	parameter count	train seconds
baseline	LSTM	20	2.2569	0.4309	9.5538	102110	7.2955
baseline	GRU	20	2.2815	0.4171	9.7915	79326	6.6768
baseline	LSTM	30	2.1888	0.4326	8.9249	102110	6.5908
baseline	GRU	30	2.2432	0.4251	9.4238	79326	6.6682
small_hidden	LSTM	30	2.2233	0.4177	9.2379	31102	6.6238
small_hidden	GRU	30	2.2773	0.4066	9.7502	24830	6.5001
wide_hidden	LSTM	30	2.2650	0.4349	9.6316	214078	25.7780
wide_hidden	GRU	30	2.3399	0.4289	10.3807	164542	13.0968
two_layers	LSTM	30	2.1718	0.4300	8.7737	234206	10.5099
two_layers	GRU	30	2.2060	0.4232	9.0791	178398	7.9328
seq50	LSTM	50	2.2217	0.4348	9.2232	102110	8.4912
seq50	GRU	50	2.2456	0.4240	9.4465	79326	7.4946

Problem 3 English-to-French Transformer Results

num layers	num heads	train loss	valid loss	sequence accuracy	bleu4	valid perplexity	parameter count	train seconds
1	2	5.8152	5.7830	0.0000	0.0000	324.7199	213780	0.2958
1	4	5.8292	5.8052	0.0000	0.0007	332.0164	213780	0.3025
2	2	5.8709	5.8159	0.0000	0.0000	335.5803	279828	0.4793
2	4	5.7950	5.8101	0.0000	0.0067	333.6489	279828	0.5159
4	2	5.8927	5.7989	0.0000	0.0000	329.9263	411924	0.8573
4	4	5.8766	5.8686	0.0000	0.0000	353.7564	411924	0.9187

Problem 4 French-to-English Transformer Results

num layers	num heads	train loss	valid loss	sequence accuracy	bleu4	valid perplexity	parameter count	train seconds
1	2	5.7091	5.5719	0.0000	0.0064	262.9208	209125	0.2811
1	4	5.6770	5.6379	0.0000	0.0016	280.8632	209125	0.2902
2	2	5.6332	5.4851	0.0000	0.0118	241.0733	275173	0.4640
2	4	5.6606	5.5904	0.0000	0.0064	267.8550	275173	0.4931
4	2	5.6468	5.5771	0.0000	0.0064	264.2971	407269	0.8348
4	4	5.6762	5.5745	0.0000	0.0075	263.6200	407269	0.9011

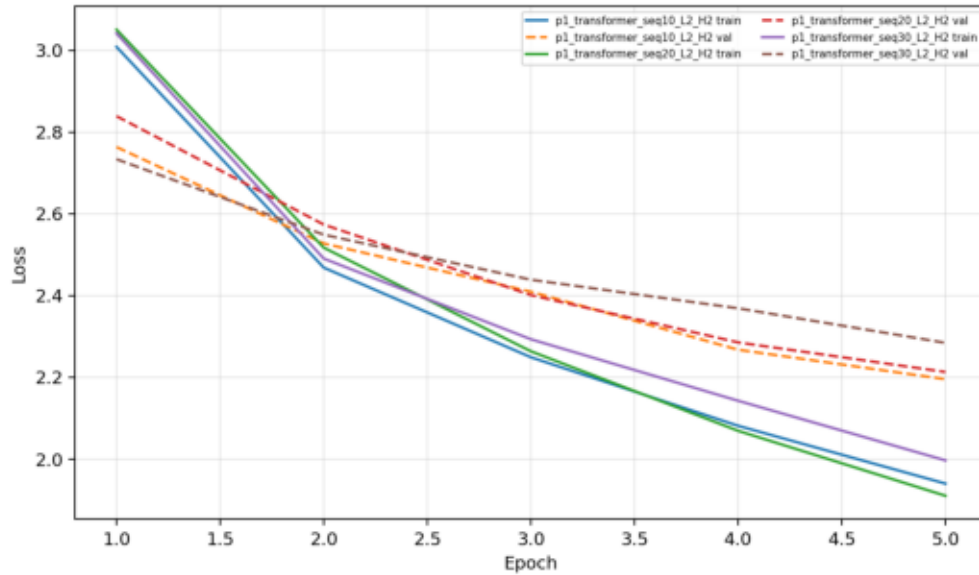
Homework 3 RNN Translation Baselines

direction	architecture	best val loss	exact match	word bleu4	char bleu4	seconds
English-to-French	Baseline GRU encoder-decode	5.2227	0.0000	0.0276	0.1463	17.5986
English-to-French	GRU with Luong attention	5.0146	0.0000	0.0589	0.2431	23.2358
French-to-English	Baseline GRU encoder-decode	4.8302	0.0000	0.0249	0.1414	14.8952
French-to-English	GRU with Luong attention	4.6855	0.0000	0.0771	0.2481	12.8894

Loss Curves

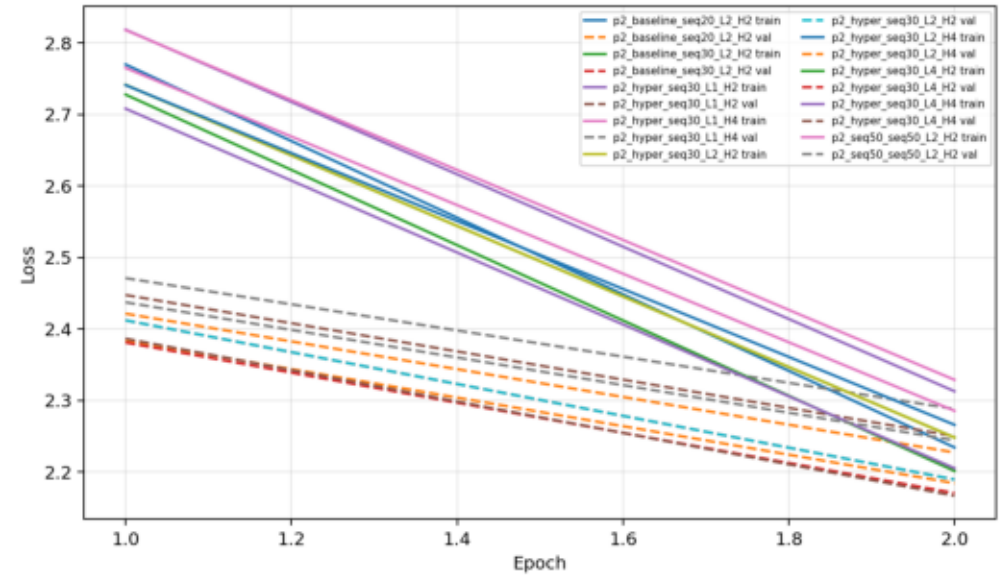
Problem1 Transformer Loss

Problem 1 Transformer Loss Curves



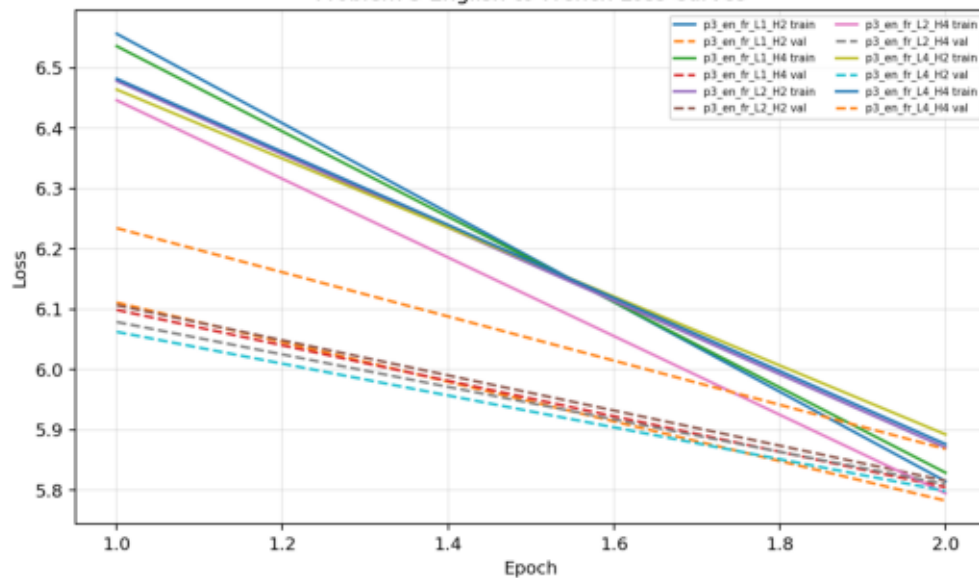
Problem2 Transformer Loss

Problem 2 Transformer Loss Curves



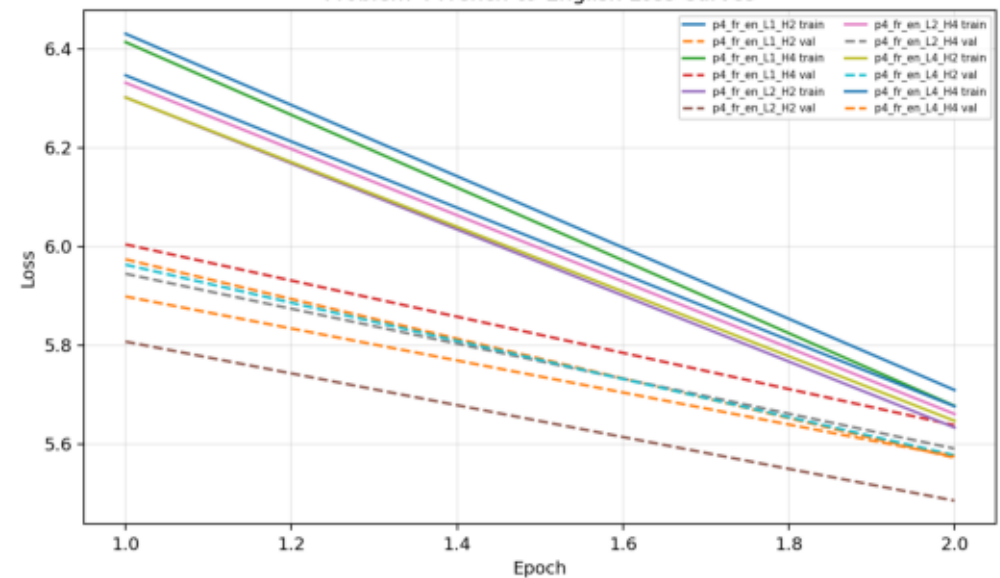
Problem3 Transformer Loss

Problem 3 English-to-French Loss Curves



Problem4 Transformer Loss

Problem 4 French-to-English Loss Curves



BLEU Comparison Plots

